DIGITAL ELECTRONICS

ET 8301

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Lecture 07

Sequential logic circuits

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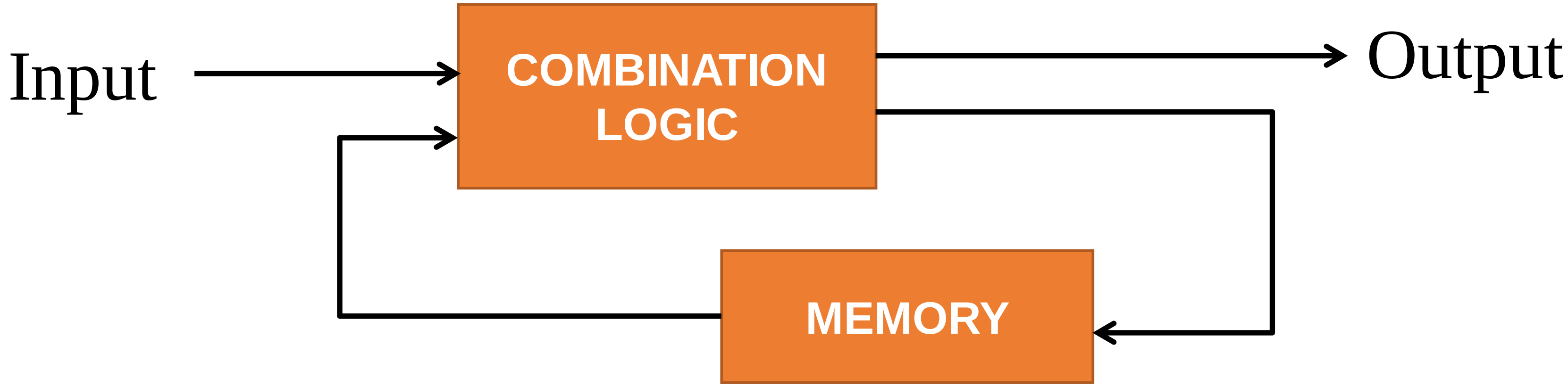
Introduction

- Engineers classify logic circuits into two groups. We have already worked with **Combination logic circuits** using **AND**, **OR**, and **NOT** gates.
- The other group of circuits is classified as **Sequential logic circuits**. Sequential circuits involve **timing** and **memory devices**.
- The basic building block for combinational logic circuits is the **logic gate**.
- The basic building block for sequential logic circuits is the **flip-flop (FF).**

Introduction

- The logic circuits whose outputs at any instant of time depend on the present inputs as well as on the past outputs are called ***sequential circuits***.
- In sequential circuits, the output signals are **feedback to the input side**.
- A block diagram of a sequential circuit is shown in



Introduction

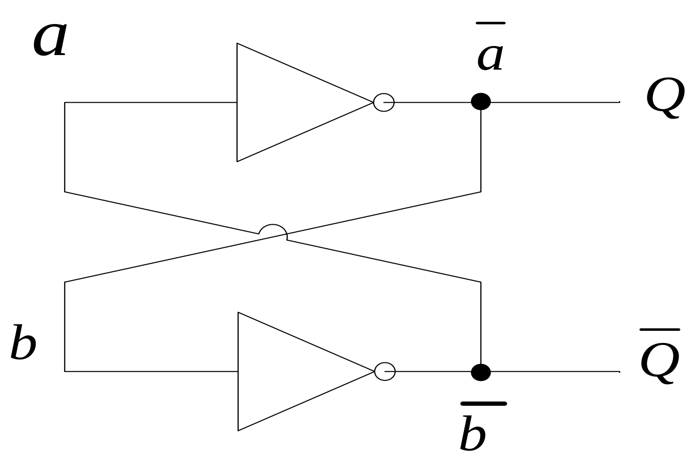
- Sequential circuits are classified broadly into **synchronous sequential circuits** and **asynchronous sequential circuits**.
- Synchronous sequential circuits are controlled by a **master clock**.
- While the operation of asynchronous sequential circuits are **not controlled by any clock pulses**. The **output change immediately to a change in input**.

Introduction

- Flip-flop is the **basic memory element in sequential circuits**.
- A flip-flop can store a **1** or a **0** indefinitely.
- We are going covers several types of flip-flop circuits.
- Also will **wire flip-flops** together. Flip-flops are wired to form other sequential circuits such as **counters**, **shift registers**, and **various memory** devices.

Basic Bistable Element

- A circuit which indefinitely store a 1 or 0 is called a ***basic bistable element***.



Basic Bistable Element

- A **flip-flop** is a bistable circuit with input lines.
- The flip-flop remains in one of its two stable states until power is **switched off** or **until an input signal triggers** a change in state.
- A flip-flop is said to be ***set or preset*** when an input forced it to store a **1** and it is said to be ***reset or cleared*** when an input signal causes the flip-flop to store a **0**.

Basic Bistable Element

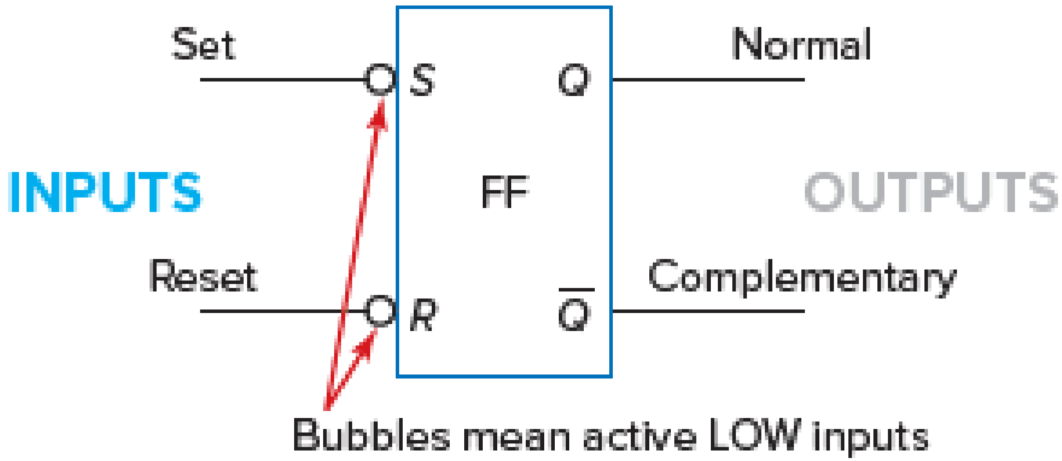
- Flip-flops are **synchronous** bistable devices, also known as **bistable multivibrators**.
- It **synchronous** because the output changes state only at a specified point (**leading or trailing edge**) on the triggering input called the **clock (CLK)** which is designated as a **control input**, C; that is, changes in the output occur in synchronization with the clock.
- Flip-flops are **edge-triggered or edge-sensitive**.

Basic Bistable Element

- A **latch** is a class of flip-flops whose output responds **immediately** to appropriate changes in the input in **spite of the presence of an enable or clock input**(asynchronous bistable devices).
- There are several types of latches based on types of **inputs** and **enable** lines.
- Latch are similar to flip-flops because they are bistable devices.
- The **main difference** between latches and flip-flops is in the **method used for changing state**.

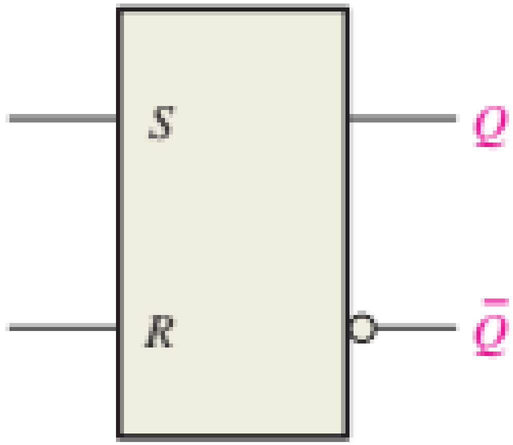
The R-S Flip-Flop

- The logic symbol for the R-S flip-flop is shown below

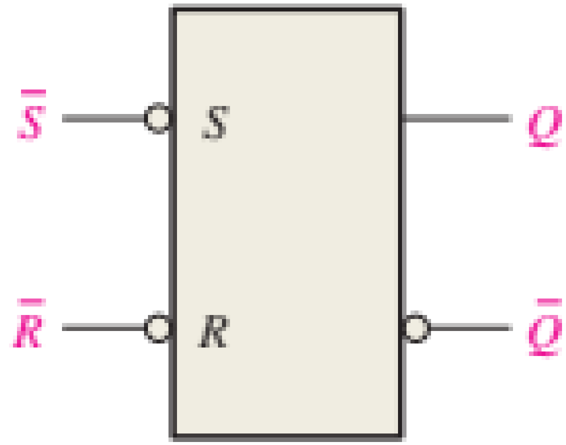


- Notice that the **R-S** flip-flop has **two inputs**, labeled **S** and **R**. The **two outputs** are labeled Q and $\overline{Q}$ (say "not Q" or "Q not").
- In flip-flops the outputs are **always opposite**, or **complementary**. In other words, if output Q = 1, then output $\overline{Q}$ = 0, and so on.

The R-S Flip-Flop



Active-HIGH input S-R Flip-Flop



Active-LOW input S-R Flip-Flop

The R-S Flip-Flop

- The R-S flip-flop symbol shown above labels the outputs as **normal** and **complementary**.
- The letters S and R at the left of the R-S flip-flop symbol are often referred to as the **set** and **reset** inputs.
- The inputs are active LOW (see bubble).
- The **R-S flip-flop** may also be referred to as an **R-S latch**.
- The term "**latch**" refers to its use as a temporary memory device. A latch such as the R-S flip-flop in Figure above can **hold one bit of information**.

The R-S Flip-Flop

- The truth table for **active low** R-S flip-flop with details the operation are shown below

Truth Table A

Mode of operation	Inputs S	R	Outputs Q	$\overline{Q}$	Effect on output Q
Prohibited	0	0	1	1	Prohibited: Do not use
Set	0	1	1	0	For setting Q to 1
Reset	1	0	0	1	For resetting Q to 0
Hold	1	1	Q	$\overline{Q}$	Depends on previous state

OR

Truth Table B

Inputs $\overline{S}$	$\overline{R}$	Outputs Q	$\overline{Q}$	Comments
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1	1	Invalid condition

The R-S Flip-Flop

- From table A when the S and R inputs are **both 0**, both outputs go to a logical **1**.This is called a **prohibited state** for the flip-flop and is **not used**.
- The second line of the truth table shows that when input **S is 0** and **R is 1**, the Q output is **set** to logical **1**. This is called the **set condition**.
- The third line shows that when input **R is 0** and **S is 1**, output Q is **reset** (cleared) to **0**. This is called the **reset condition**.
- The fourth line in the truth table shows both **inputs (R and S) at 1**. This is the **idle or at rest condition** and leaves Q and $\overline{Q}$ in their previous complementary states. This is called the **hold condition**.

The R-S Flip-Flop

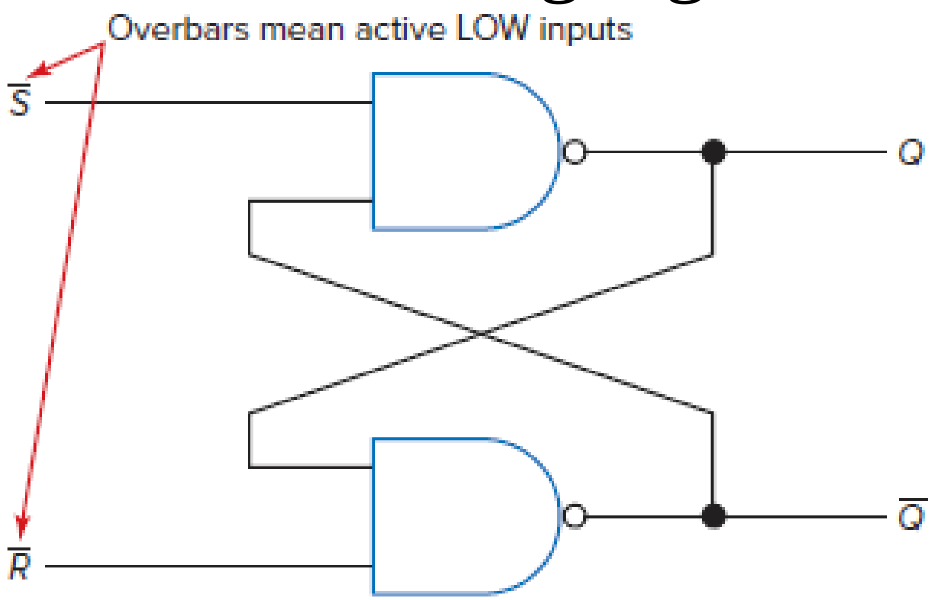
- From truth table , it may be observed that it takes a **logical 0 to activate the set** (set Q to 1).
- It also takes a **logical 0 to activate the reset**, or clear (clear Q to 0).
- Because it takes **a logical 0 to enable, or activate**, the flip-flop, the logic symbol in figure above is active low flipflop and has **invert bubbles at the R and S** inputs.
- These invert bubbles indicate that the set and reset inputs are activated by a logical 0.

The R-S Flip-Flop

- From truth table , it may be observed that it takes a **logical 0 to activate the set** (set Q to 1).
- It also takes a **logical 0 to activate the reset**, or clear (clear Q to 0).
- Because it takes **a logical 0 to enable, or activate**, the flip-flop, the logic symbol in figure above is active low flipflop and has **invert bubbles at the R and S** inputs.
- These invert bubbles indicate that the set and reset inputs are activated by a logical 0.

The R-S Flip-Flop

- R-S flip-flops can be purchased in an IC package, or they can be wired from logic gates, as shown in figure below

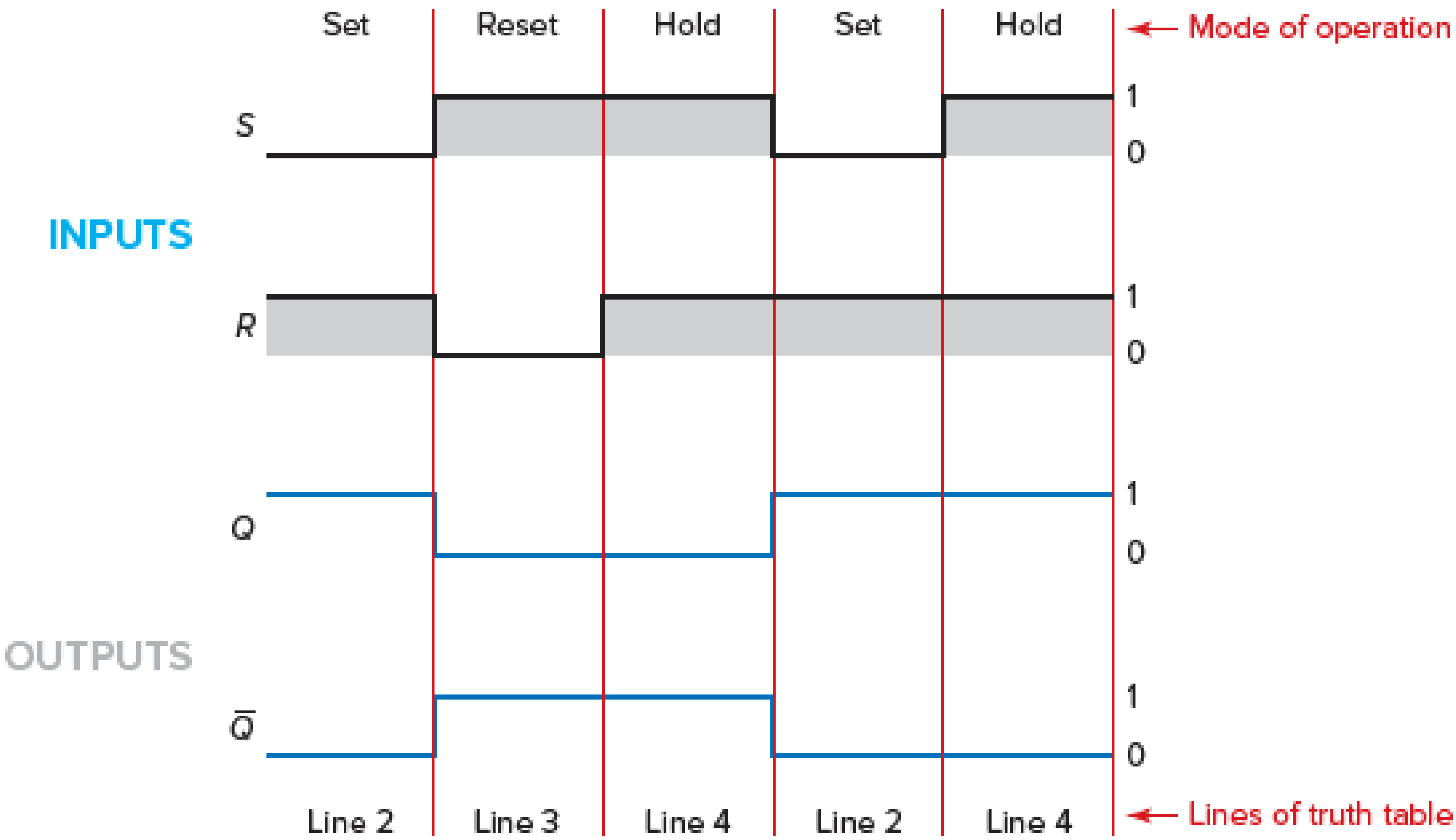


- The NAND gates in figure above form an R-S flip-flop. This NAND-gate R-S flip-flop operates according to the truth table.

The R-S Flip-Flop

- Technically, the R-S flip-flop in figure above might be referred to as an **$\overline{R}$-$\overline{S}$ flipflop** or **$\overline{R}$-$\overline{S}$ latch**.
- The overbars above the R and S mean these are **active LOW inputs**.
- Many times **timing diagrams, or waveforms**, are given for sequential logic circuits.
- These diagrams show the voltage level and timing between inputs and outputs and are similar to what you would observe on an **oscilloscope**.
- The horizontal distance is **time**, and the vertical distance is **voltage**.

The R-S Flip-Flop



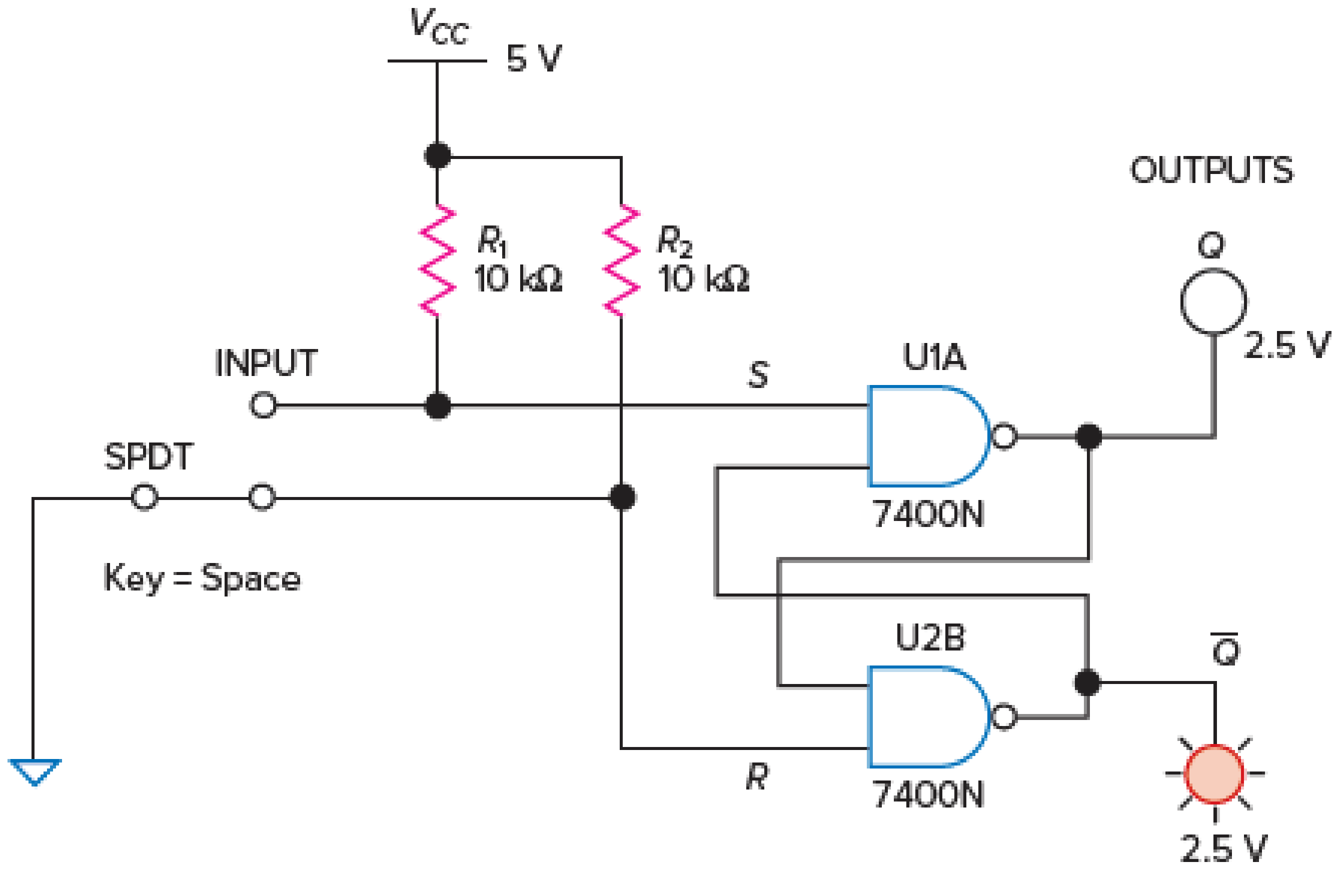
The R-S Flip-Flop

- There are three types of **multivibrators** (MVs). They are **the monostable MV**, **the astable MV**, and **the bistable multivibrator**.
- The R-S flip-flop is one of several bistable MVs. The R-S flip-flop is most commonly known as a latch and is listed under this heading in IC latch catalogs.

The R-S Flip-Flop

- A **latch** is a fundamental binary memory device for holding data.
- Latches are ***commonly used at the output of a digital device to hold the data until the next device is ready to receive the input.***
- Latches are commonly organized into groups of 4-bits, 8-bits, or more into registers.
- An 8-bit register would be a group of eight latches holding a **byte of information**.
- R-S flip-flops were also used for switch debouncing.

The R-S Flip-Flop



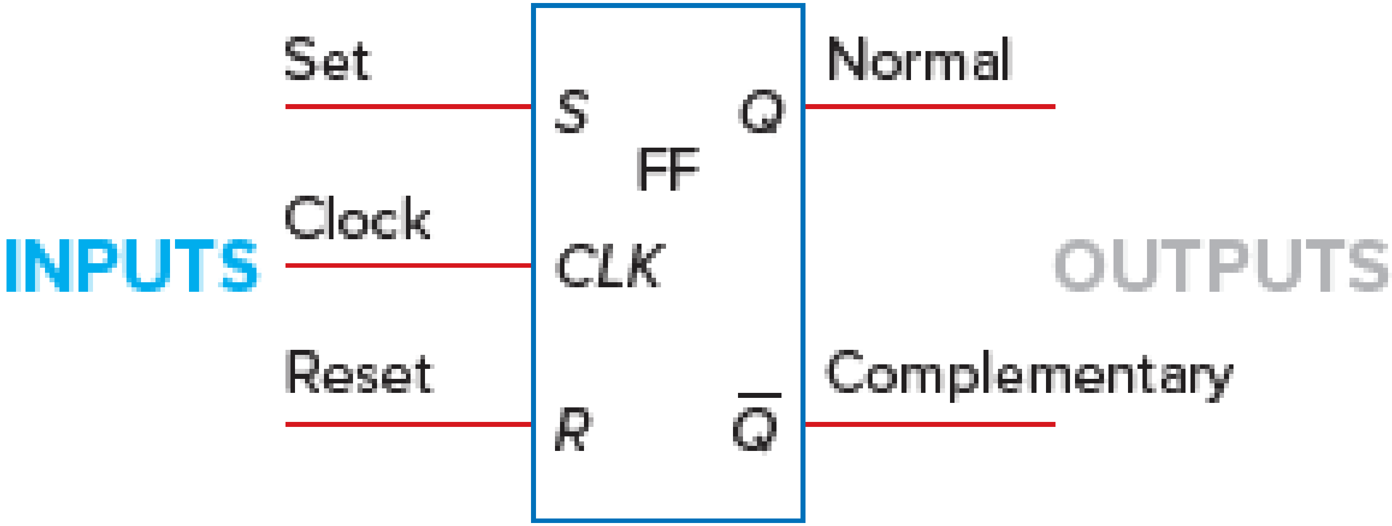
- ➢R-S flip-flop in an SPDT switch debouncing circuit.

The R-S Flip-Flop

- The logic symbol and truth table for the R-S flip-flop?
- The four modes of operation for the R-S flip-flop?

The Clocked R-S Flip-Flop

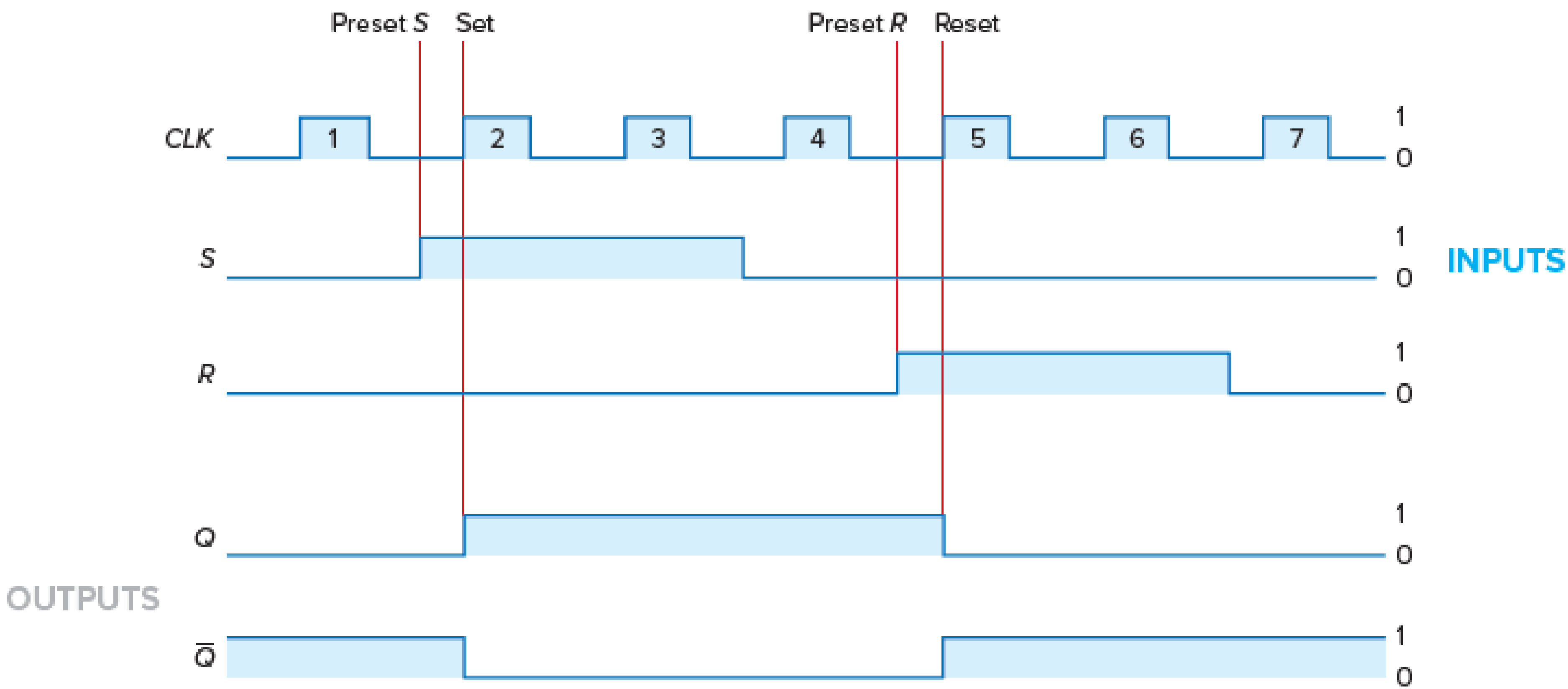
➢The logic symbol for a **clocked R-S flip-flop** is shown in figure below.



➢You can observe that it looks almost like an R-S flip-flop except that **it has one extra input labeled CLK (for clock)**.

The Clocked R-S Flip-Flop

- ➢The operation of the clocked R-S flip-flop is detailed in figure below where the CLK input is at the top of the diagram as shown below.



The Clocked R-S Flip-Flop

- Notice that the **clock pulse (1)** has no effect on output Q with inputs S and R in the 0 position(Active High). The flip-flop is in the idle, or hold, mode during clock pulse 1.
- At the preset S position, the S (set) input is moved to 1, but **output Q is not yet set to 1**.
- The rising edge of **clock pulse 2** permits **Q to go to 1**. Pulses 3 and 4 have no effect on output Q. During pulse 3, the flip-flop is in its set mode, and during pulse 4, it is in its hold mode.

The Clocked R-S Flip-Flop

- Next, **input R is preset to 1**. On the rising edge of clock pulse 5, **the Q output is reset (or cleared) to 0**.
- The flip-flop is in the reset mode during both clock pulses 5 and 6.
- The flip-flop is in its hold mode during **clock pulse 7**; therefore, the normal output (Q) remains at **0**.

The Clocked R-S Flip-Flop

- Notice that the outputs of the clocked R-S flip-flop change **only on a clock pulse**.
- We say that this flip-flop **operates synchronously**; it operates in step with the clock.
- Synchronous operation is very important in most digital circuits, where each step must happen in an exact order.
- Another characteristic of the clocked R-S flip-flop is that once it is set or reset, **it stays that way even if you change some inputs**.
- This is a **memory characteristic**, which is extremely valuable in many digital circuits.

The Clocked R-S Flip-Flop

- The truth table for the Clocked R-S Flip-Flop.

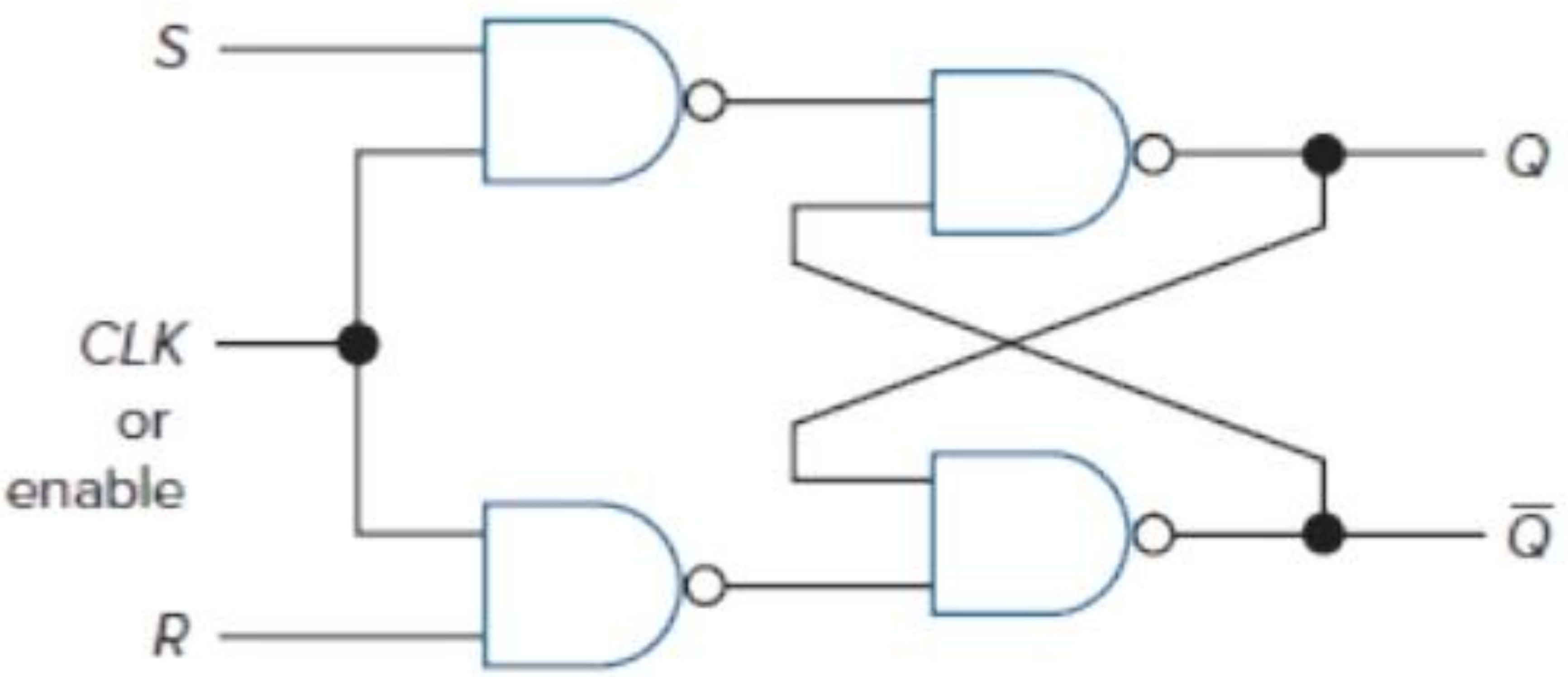
Mode of operation	INPUTS			OUTPUTS		
	CLK	*S*	*R*	*Q*	$\overline{Q}$	Effect on output *Q*
Hold		0	0	No change		No change
Reset		0	1	0	1	Reset or cleared to 0
Set		1	0	1	0	Set to 1
Prohibited		1	1	1	1	Prohibited—do not use

The Clocked R-S Flip-Flop

- From the truth table above for the clocked R-S flip-flop. You can notice that only the **top three lines** of the truth table are usable; **the bottom line is prohibited and not used**.
- You can also observe that the R and S inputs to the clocked R-S flip-flop are **active HIGH inputs**.
- That is, it takes a **HIGH** on input **S** while **R = 0** to cause output **Q to be set to 1**.

The Clocked R-S Flip-Flop

- The **wiring diagram** of a clocked R-S flip-flop

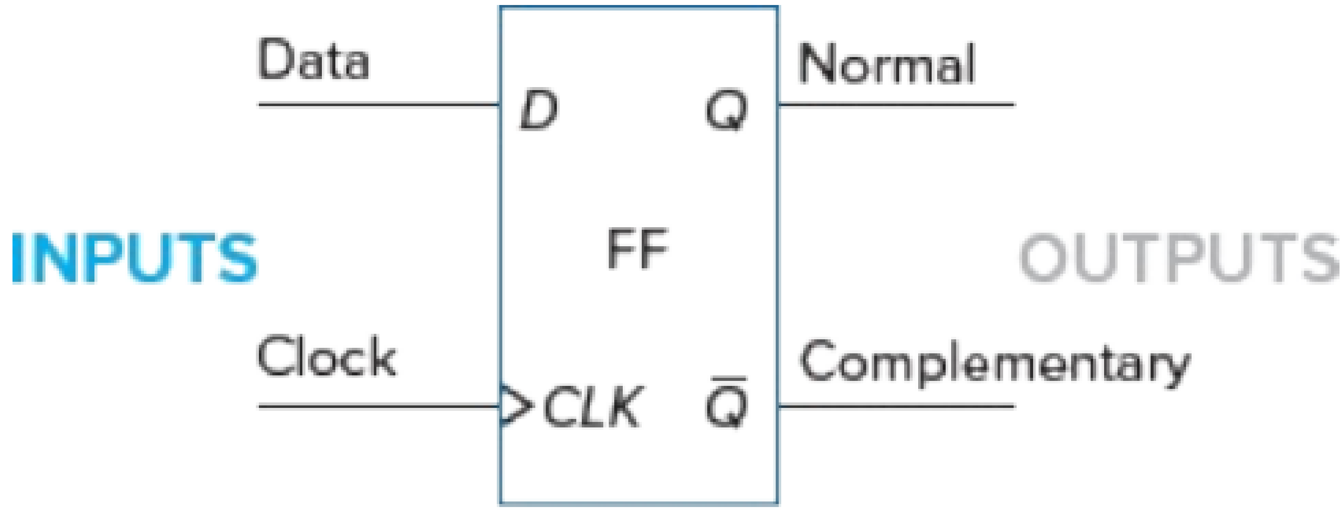


The Clocked R-S Flip-Flop

- You can notice that two **NAND gates** have been **added to the inputs** of the R-S flip-flop to add the clocked feature.
- The CLK input may be labeled with a **C or E** for enable by various manufacturers.
- It is **important to remember** that the memory characteristics exhibited by flip-flops are among the fundamental reasons why digital technology has become so widely used in modern electronic products.

The D Flip-Flop

- The logic symbol for the D flip-flop is shown in figure below.



- It has only **one data input (D)** and a **clock input (CLK)**. The **outputs** are labeled **Q** and **$\overline{\text{Q}}$**.

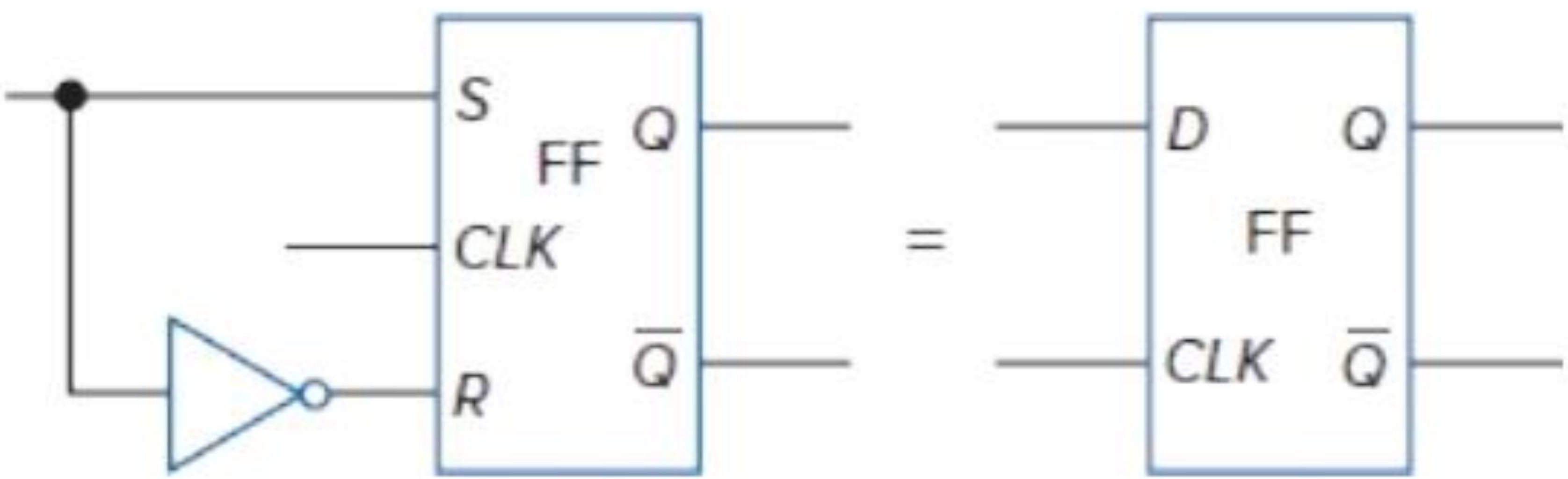
The D Flip-Flop

- The D flip-flop is often called a **delay flip-flop**.
- The word delay describes **what happens to the data, or information**, at input D.
- The data (a **0 or 1**) at input D is **delayed one clock pulse** from getting to output Q.
- The truth table for the D flip-flop is shown below

Input	Output
D	Q^{n+1}
0	0
1	1

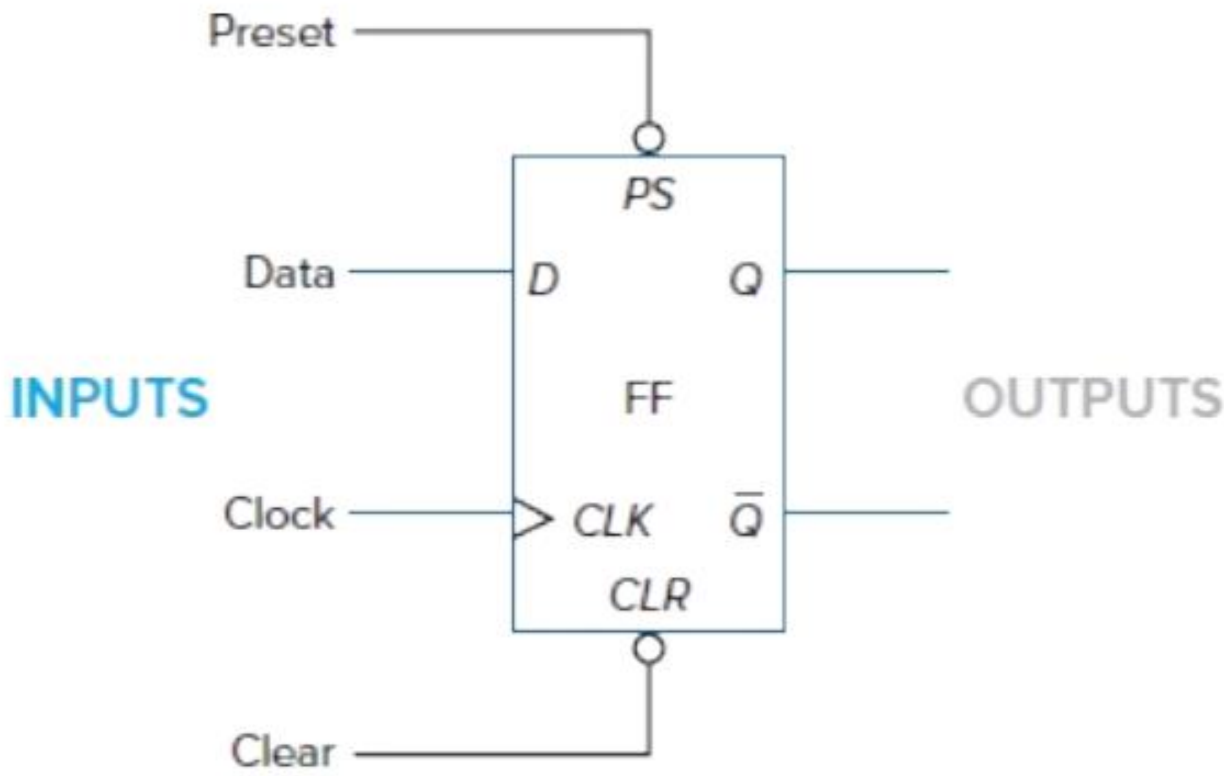
The D Flip-Flop

- Notice that output Q follows **input D after one clock pulse** (see Q^{n+1} column).
- A **D flip-flop may be formed from a clocked R-S flip-flop** by adding an inverter, as shown in figure below



The D Flip-Flop

- More commonly we will use a D flip-flop contained in an IC. The figure below shows a typical commercial D flip-flop.



The D Flip-Flop

- Two extra inputs [**PS (preset)** and **CLR (clear)**] have been added to the D flip-flop in Figure above
- The **PS input** sets **output Q to 1** when enabled by a logical **0**. The CLR **input clears output Q to 0** when enabled by a logical **0**.
- The **PS** and **CLR** inputs will **override** the D and CLK inputs.
- Note the addition of a **small triangle** on the CLK input of the IC symbol means the flip-flop is **edge-triggered**.

The D Flip-Flop

- During synchronous operation, edge triggering means the bit of data present at input D is transferred to output Q on the **positive edge (LOW-to-HIGH transition)** of the clock pulse.
- The **7474 D flip-flop IC** is said to be positive-edge triggered.

The D Flip-Flop

- The truth table for 7474 D flip-flop.

Mode of operation	INPUTS				OUTPUTS	
	Asynchronous		Synchronous			
	PS	*CLR*	*CLK*	*D*	*Q*	$\overline{Q}$
Asynchronous set	0	1	X	X	1	0
Asynchronous reset	1	0	X	X	0	1
Prohibited	0	0	X	X	1	1
Set	1	1	↑	1	1	0
Reset	1	1	↑	0	0	1

0 = LOW
1 = HIGH
X = Irrelevant
↑ = LOW-to-HIGH transition of clock pulse

The D Flip-Flop

- **Remember** that the asynchronous (not synchronous) inputs (PS and CLR) **override the synchronous inputs**.
- The asynchronous inputs are in control of the D flip-flop in the first three lines of the truth table above.
- The synchronous inputs (D and CLK) are irrelevant as shown by the **X's** on the truth table.
- The **prohibited condition, line 3** on the truth table, should be **avoided**.
- With **both asynchronous inputs disabled** (PS = 1 and CLR = 1), the D flip-flop can be set and reset using the **D** and **CLK** inputs.

The D Flip-Flop

- The last two lines of the truth table use a clock pulse to transfer data from **input D to output Q** of the flip-flop. Being in step with the clock, this is called **synchronous operation**.
- **Note** that this flipflop uses the **LOW-to-HIGH** transition of the clock pulse to **transfer data from input D to output Q**.

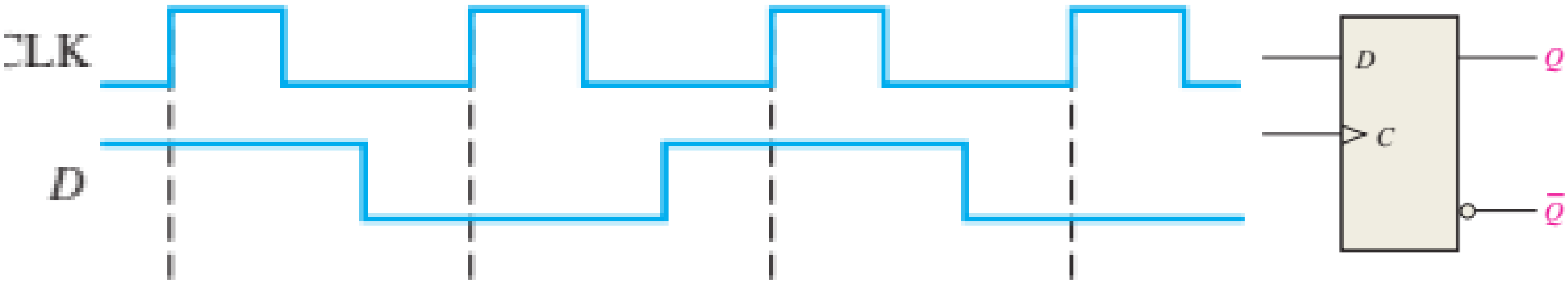
The D Flip-Flop

- **D flip-flops** are sequential logic devices, which are widely used **temporary memory devices**.
- D flip-flops are wired together to form **shift registers** and **storage registers**. These registers are commonly used in digital systems.
- **Remember** that the D flip-flop **delays the input data from reaching the output** (Q) until a clock pulse has taken place.
- Due to this delay, it is called a **delay flip-flop**. D flip- flops are sometimes also called **data flipflops** or **D-type latches**.
- D flip-flops are available in both **TTL** and **CMOS** IC form.

The D Flip-Flop

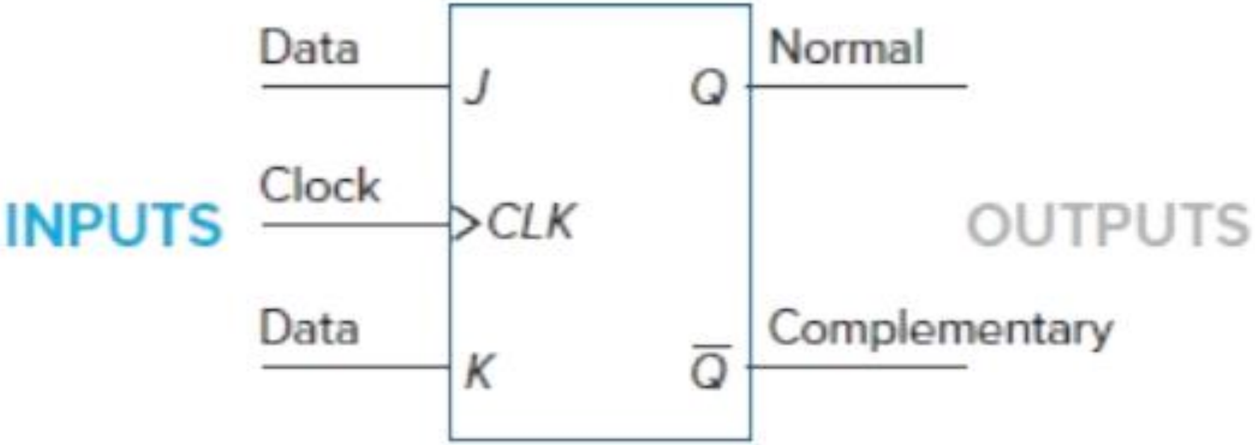
- **Example**

Given the waveforms in Figure below for the ***D* input** and the clock, determine the $\boldsymbol{Q}$ ***output*** and $\overline{\boldsymbol{Q}}$ ***output*** waveform if the flip-flop starts out RESET.



The J-K Flip-Flop

- The J-K flip-flop has the features of **all the other types of flip-flops**.
- The logic symbol for the J-K flip-flop is illustrated in figure below.



- The inputs labeled **J** and **K** are **the data inputs**. The input labeled **CLK** is the **clock input**.
- **Outputs Q** and $\overline{Q}$ are the usual **normal** and **complementary** outputs on a flip-flop.

The J-K Flip-Flop

- A truth table for the J-K flip-flop is shown below

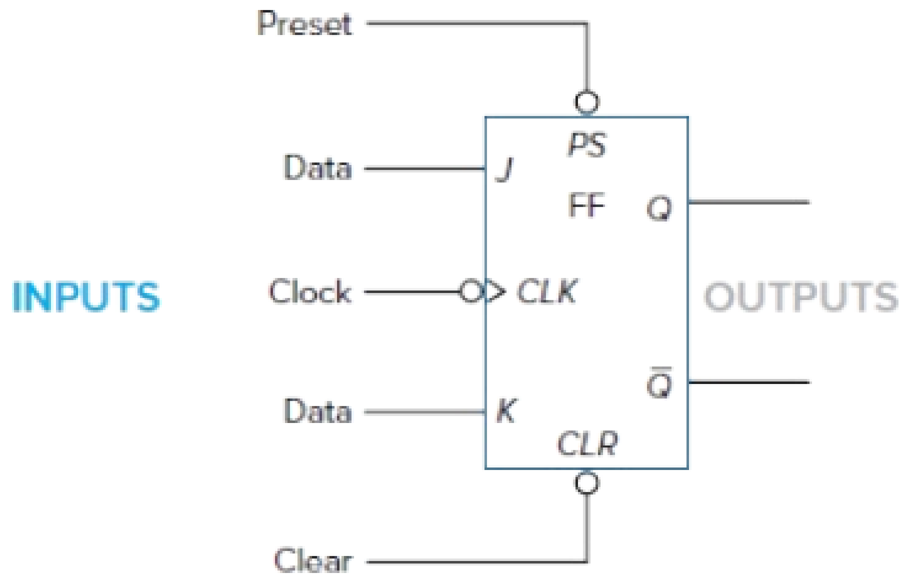
Mode of operation	INPUTS			OUTPUTS		
	CLK	J	K	Q	$\overline{Q}$	Effect on output Q
Hold		0	0	No change		No change—disable
Reset		0	1	0	1	Reset or cleared to 0
Set		1	0	1	0	Set to 1
Toggle		1	1	Toggle		Changes to opposite state

The J-K Flip-Flop

- When the **J** and **K** inputs are both **0**, the flip-flop is in the **hold mode**.
- In the hold mode, **the data inputs have no effect on the outputs**. The outputs "**hold**" the last data present.
- From the truth table **lines 2** and **3** show that the **reset** and **set** conditions for the **Q output**.
- **Line 4** illustrates the useful **toggle position of the J-K flip-flop**.
- When both data inputs **J** and **K** are at **1**, repeated clock pulses cause the output to **turn off-on-off-on-off-on**, and so forth.
- This off-on action is like a toggle switch and is called **toggling**.

The J-K Flip-Flop

- The logic symbol for the commercial **7476 TTL J-K flip-flop** is shown in figure below



- The ***two asynchronous inputs*** were added to the symbol two (**preset** and **clear**).
- The **synchronous inputs** are the J and K data and clock inputs.

The J-K Flip-Flop

- The **customary normal (Q)** and **complementary $\overline{Q}$** outputs. A detailed truth table for the commercial 7476 J-K flip-flop is shown in figure below

Mode of operation	INPUTS					OUTPUTS	
	Asynchronous		Synchronous				
	PS	*CLR*	*CLK*	*J*	*K*	*Q*	$\overline{Q}$
Asynchronous set	0	1	X	X	X	1	0
Asynchronous reset	1	0	X	X	X	0	1
Prohibited	0	0	X	X	X	1	1
Hold	1	1	⎍	0	0	No change	
Reset	1	1	⎍	0	1	0	1
Set	1	1	⎍	1	0	1	0
Toggle	1	1	⎍	1	1	Opposite state	

0 = LOW
1 = HIGH
X = Irrelevant
⎍ = Positive clock pulse

The J-K Flip-Flop

- You can recall the **asynchronous inputs** (such as **PS** and **CLR**) from the D Flipflop section **always will override synchronous inputs**.
- The asynchronous inputs are **activated in the first three lines** of the truth table.
- The synchronous inputs are irrelevant (overridden) in the first three lines as shown in the truth table above.
- The **prohibited state** occurs when both asynchronous inputs are activated at the same time. The prohibited state is **not useful** and should be **avoided**.

The J-K Flip-Flop

- When both **asynchronous inputs** (**PS** and **CLR**) are **disabled** with a 1, the synchronous inputs can be **activated**.
- The bottom four lines of the truth table above detail the **hold**, **reset**, **set**, and **toggle modes** of operation for the 7476 J-K flip-flop.
- **Note that** the 7476 J-K flip-flop uses the entire pulse to **transfer** data from the **J** and **K data inputs** to the **Q** and $\mathbf{\overline{Q}}$ outputs.
- J-K flip-flops are used in many digital circuits. You will use the J-K flip-flop especially in **counters**.
- Counters are found in almost every digital system.

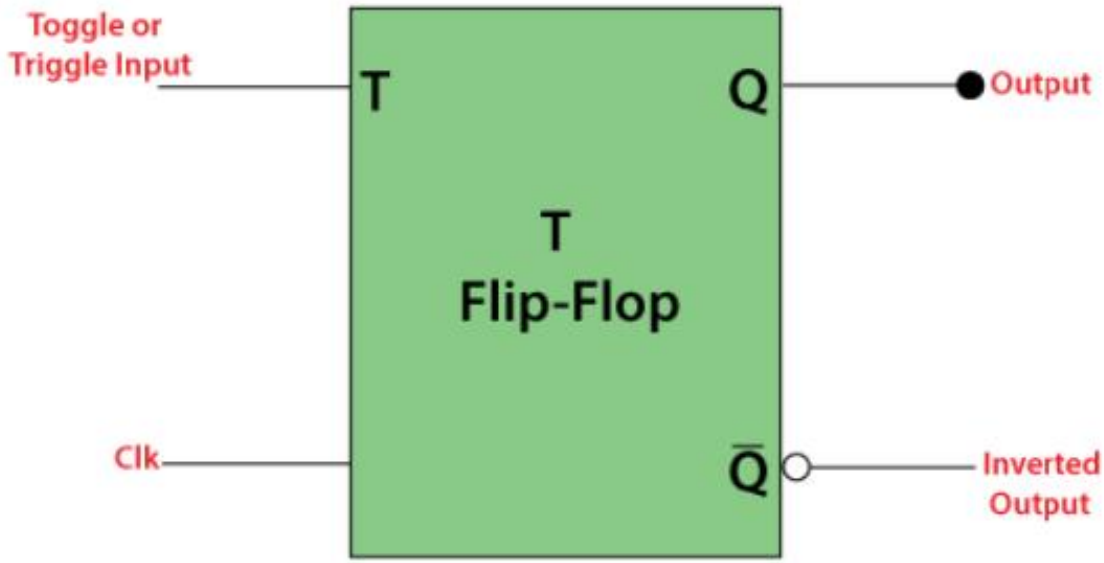
The J-K Flip-Flop

In **summary**,

- the J-K flip-flop is considered the "**universal**" **flip-flop**. Its unique feature is the toggle mode of operation so useful in designing counters.
- When the J-K flip-flop is wired for use only in the **toggle mode**, it is commonly called a **T flip-flop**.

The T Flip-Flop

- The **T flip-flop** has the features like all the other types of flip-flops.
- The logic symbol for the T flip-flop is illustrated in figure below.



The T Flip-Flop

- In T flip flop, "**T**" defines the term "**Toggle**".
- In **T Flip Flop**, we provide only a single input called "**Toggle**" or "**Trigger**" input to avoid an intermediate state occurrence.
- It work as a **Toggle switch**.
- The **next output state is changed** with the complement of the **present state output**. This process is known as "**Toggling**".

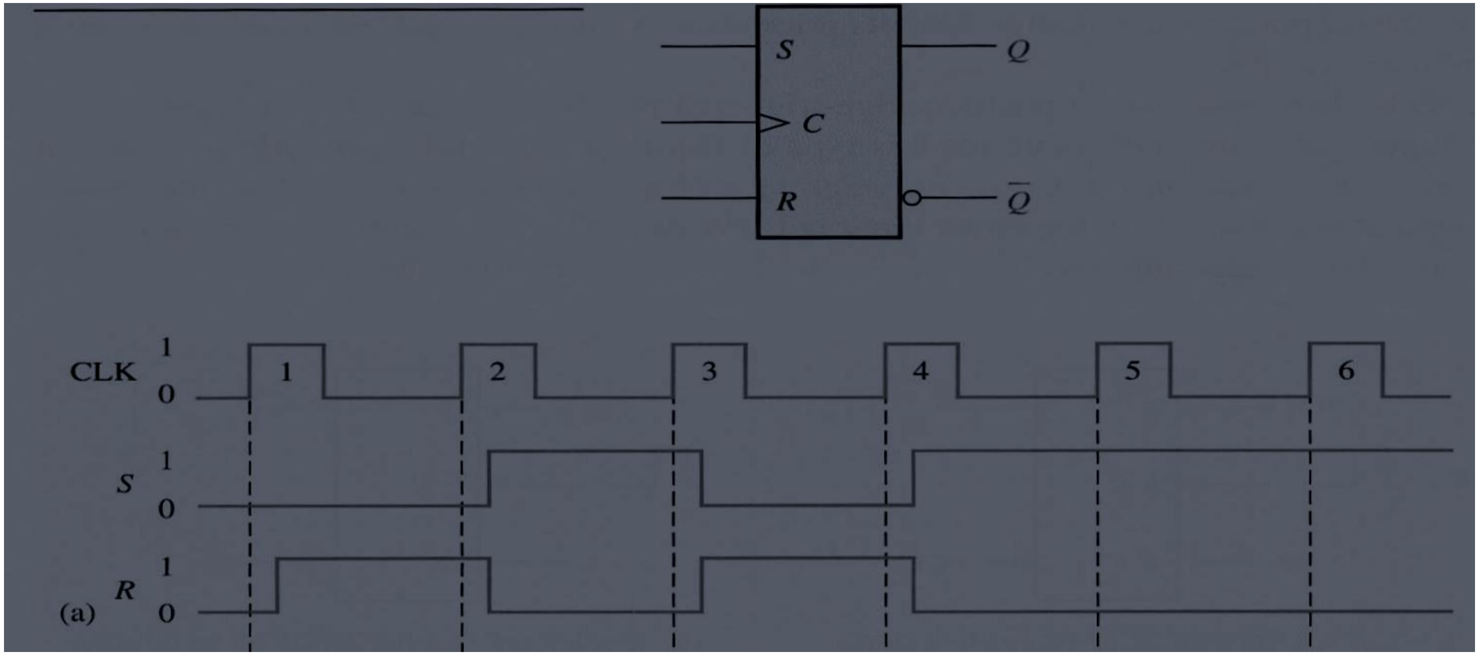
The T Flip-Flop

- Truth Table of T Flip Flop is shown in table below

	Previous		Next	
T	Q	Q'	Q	Q'
0	0	1	0	1
0	1	0	1	0
1	0	1	1	0
1	1	0	0	1

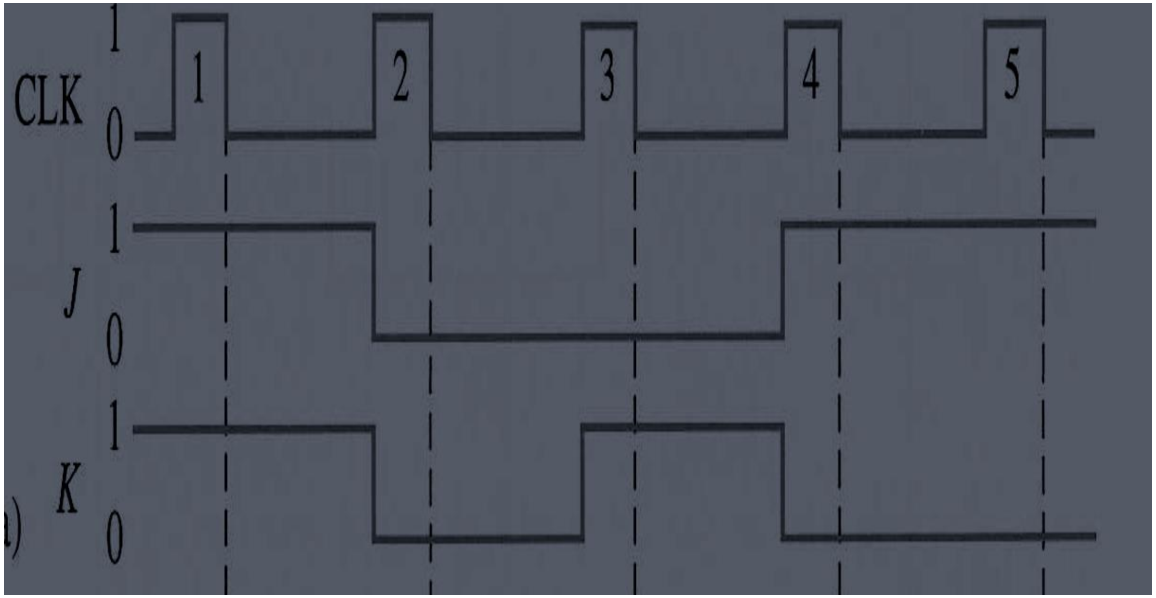
Tutorials

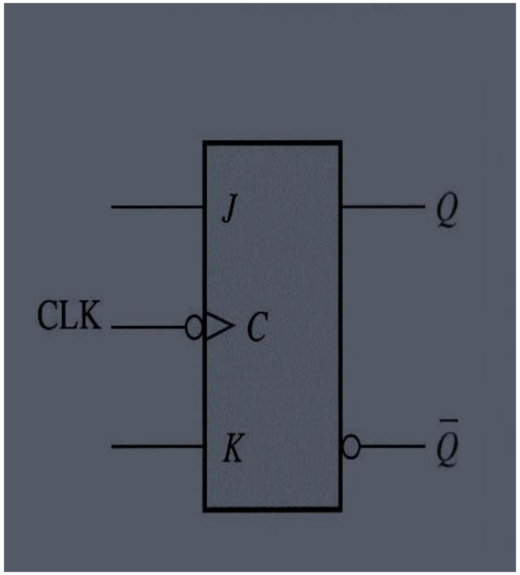
- Determine the Q and $\bar{Q}$ output waveforms of the flip-flop in Fig 1 for S, R and CLK inputs. Assume that the positive triggered flip-flop is initially RESET



Tutorials

- The waveforms of Fig are applied to the J, K and clock inputs as indicated. Determine the Q output assuming the flip-flop is initially RESET





Homework

- Read chapter 7 of Digital Fundamentals by Floyd
- Read more above Negative Edge triggering flipflop